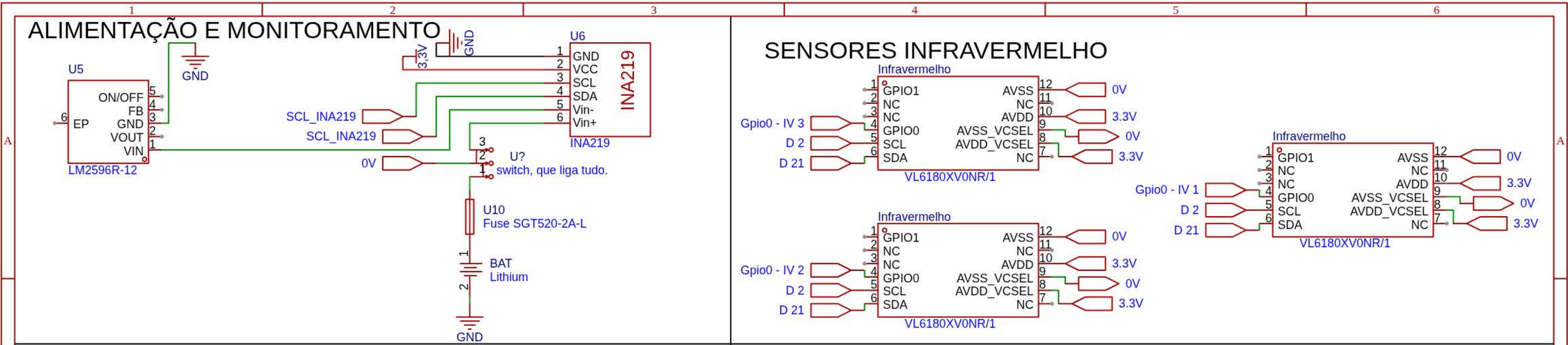




ALIMENTAÇÃO E MONITORAMENTO

U5: LM2596R-12 (Voltage Regulator)

U6: INA219 (Current Sensor)

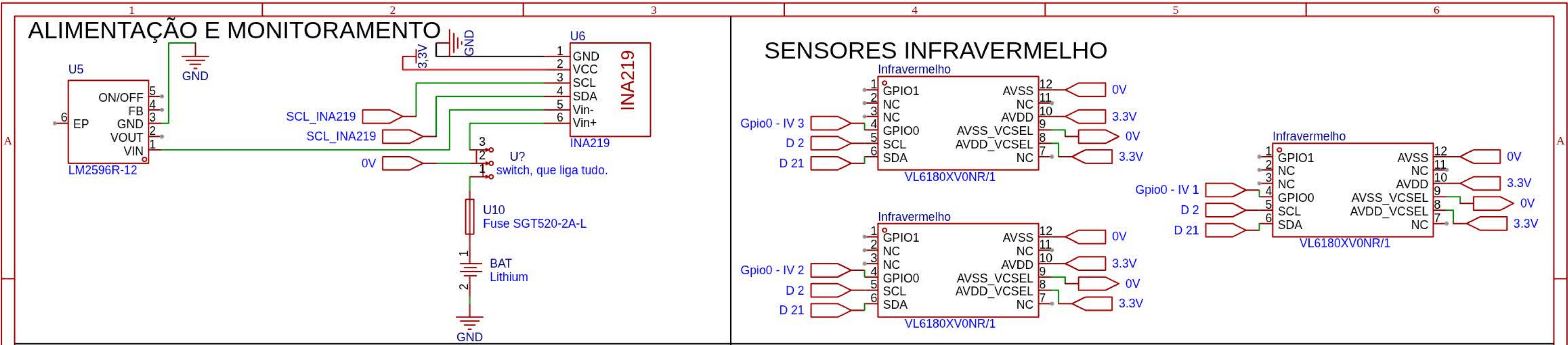
U10: Fuse SGT520-2A-L

BAT: Lithium (Battery)

U?: switch, que liga tudo.

Connections:

- U5 VIN: 3.3V
- U5 GND: GND
- U5 VOUT: U6 VCC
- U6 GND: GND
- U6 SCL: SCL_INA219
- U6 SDA: SCL_INA219
- U6 Vin-: 0V
- U6 Vin+: U?
- U10: BAT to GND



This diagram illustrates the hardware setup for controlling two N20 motors with encoders using an ESP32 microcontroller. The setup is divided into three main sections: Motor A, Motor B, and the Microcontroller.

MOTORES E CONTROLE

MOTOR A: An N20 Motor With Encoder. Its pins are connected as follows:

- MOTOR + (Pin 1) to 3.3V
- MOTOR - (Pin 6) to GND
- EN-CH1 (Pin 2) to MA_CH1
- EN-CH2 (Pin 3) to MA_CH2
- GND (EN-) (Pin 4) to GND
- int1 (Pin 10) to INT1
- int2 (Pin 9) to INT2
- int3 (Pin 8) to INT3
- int4 (Pin 7) to INT4

MOTOR B: An N20 Motor With Encoder. Its pins are connected as follows:

- MOTOR + (Pin 1) to 3.3V
- MOTOR - (Pin 6) to GND
- EN-CH1 (Pin 2) to MB_CH1
- EN-CH2 (Pin 3) to MB_CH2
- GND (EN-) (Pin 4) to GND
- int1 (Pin 10) to INT1
- int2 (Pin 9) to INT2
- int3 (Pin 8) to INT3
- int4 (Pin 7) to INT4

U9 MX1508: A motor driver module. Its pins are connected as follows:

- IN+ (Pin 5) to +5V
- IN- (Pin 6) to GND
- MA1 (Pin 7) to MOTOR A +
- MA2 (Pin 8) to MOTOR B +
- MB1 (Pin 9) to MOTOR A -
- MB2 (Pin 10) to MOTOR B -

switch que alimenta o motor: A switch connected to the +5V and GND lines of the motor driver module.

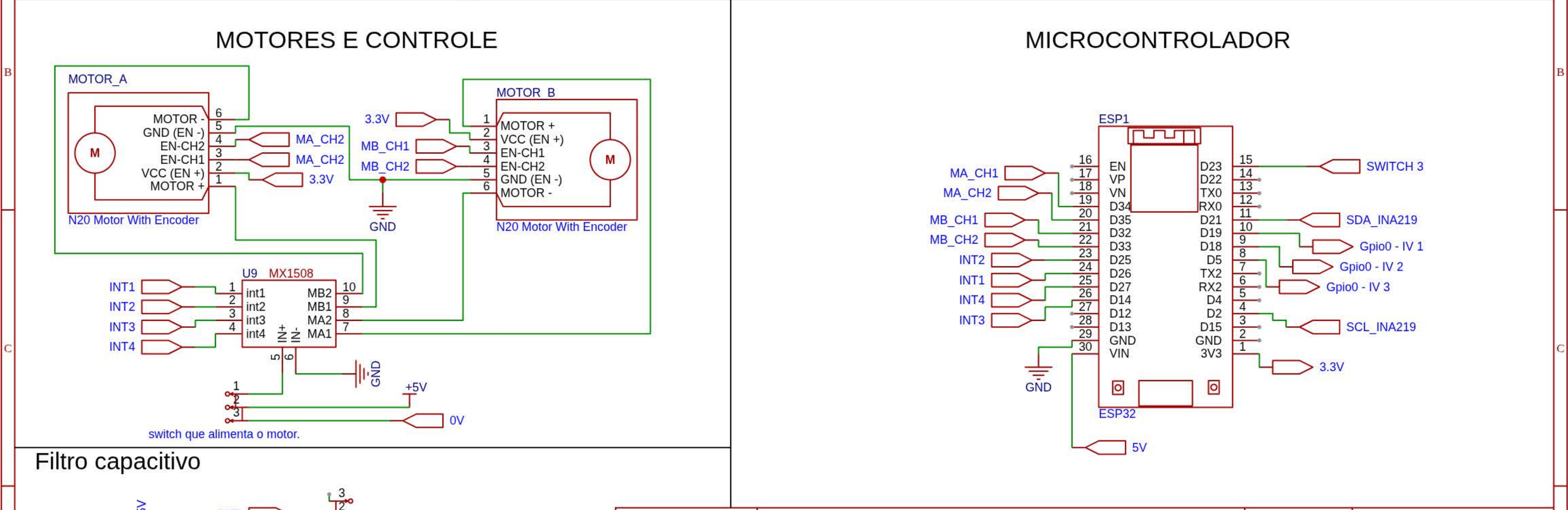
MICROCONTROLADOR

ESP1: The microcontroller module. Its pins are connected as follows:

- EN (Pin 16) to MA_CH1
- VP (Pin 17) to MA_CH2
- VN (Pin 18) to MB_CH1
- VP (Pin 19) to MB_CH2
- D34 (Pin 20) to INT2
- D35 (Pin 21) to INT1
- D32 (Pin 22) to INT4
- D33 (Pin 23) to INT3
- D25 (Pin 24) to GND
- D26 (Pin 25) to GND
- D27 (Pin 26) to GND
- D14 (Pin 27) to GND
- D12 (Pin 28) to GND
- D13 (Pin 29) to GND
- GND (Pin 30) to GND
- VIN (Pin 31) to 5V

ESP32: The microcontroller module. Its pins are connected as follows:

- D23 (Pin 15) to SWITCH 3
- D22 (Pin 14) to SWITCH 3
- TX0 (Pin 13) to SWITCH 3
- RX0 (Pin 12) to SWITCH 3
- D21 (Pin 11) to SDA_INA219
- D19 (Pin 10) to Gpio0 - IV 1
- D18 (Pin 9) to Gpio0 - IV 2
- D5 (Pin 8) to Gpio0 - IV 3
- TX2 (Pin 7) to Gpio0 - IV 3
- RX2 (Pin 6) to Gpio0 - IV 3
- D4 (Pin 5) to SCL_INA219
- D2 (Pin 4) to SCL_INA219
- D15 (Pin 3) to SCL_INA219
- GND (Pin 2) to GND
- 3V3 (Pin 1) to 3.3V



This diagram illustrates the hardware setup for controlling two N20 motors with encoders using an ESP32 microcontroller. The system is divided into three main sections: Motor Control, Microcontroller, and a note on capacitive filtering.

MOTORES E CONTROLE

MOTOR_A and MOTOR_B: Both are N20 Motor With Encoder units. Their pin connections are as follows:

- MOTOR_A:** 1 (MOTOR +) to 3.3V, 2 (VCC (EN +)) to 3.3V, 3 (EN-CH1) to MA_CH2, 4 (EN-CH2) to MA_CH1, 5 (GND (EN -)) to GND, 6 (MOTOR -) to GND.
- MOTOR_B:** 1 (MOTOR +) to 3.3V, 2 (VCC (EN +)) to 3.3V, 3 (EN-CH1) to MB_CH2, 4 (EN-CH2) to MB_CH1, 5 (GND (EN -)) to GND, 6 (MOTOR -) to GND.

U9 MX1508: A motor driver module with the following connections:

- int1 to INT1, int2 to INT2, int3 to INT3, int4 to INT4.
- MA1 to MOTOR_A pin 1, MA2 to MOTOR_A pin 2, MA3 to MOTOR_A pin 3.
- MB1 to MOTOR_B pin 1, MB2 to MOTOR_B pin 2, MB3 to MOTOR_B pin 3.
- IN+ to 5V, IN- to GND.

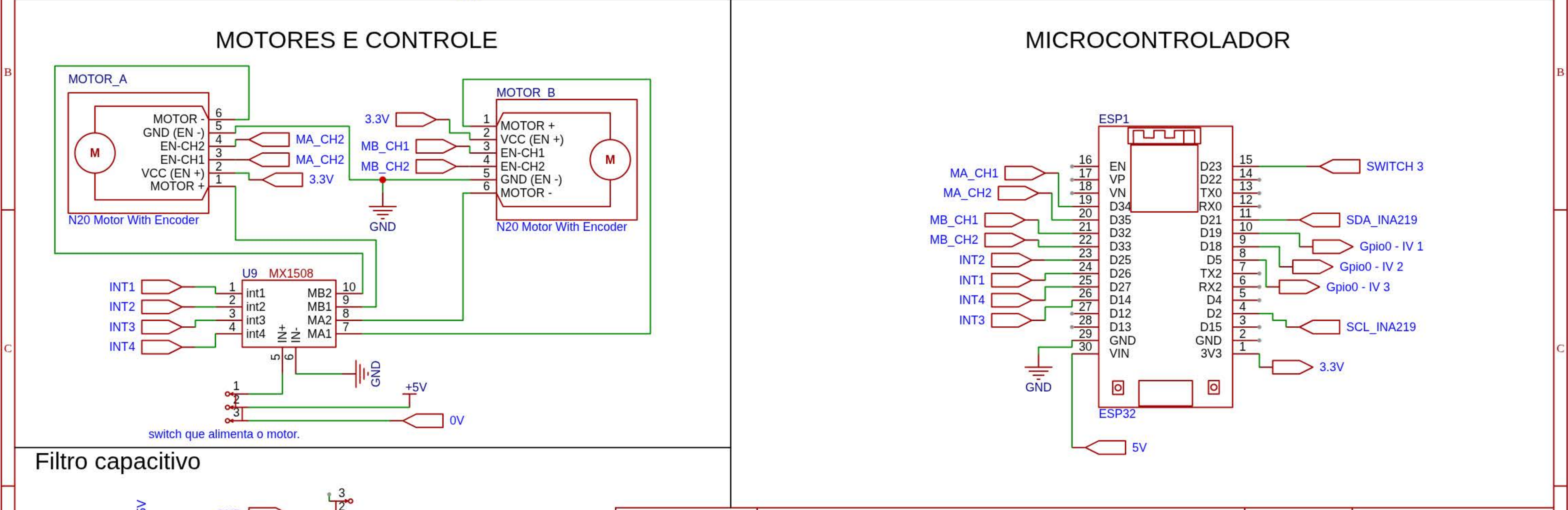
Power and Control: A switch labeled "switch que alimenta o motor." is connected to a +5V supply and ground (0V).

MICROCONTROLADOR

ESP1 and ESP32: The ESP32 microcontroller is connected to the motor control components and external devices.

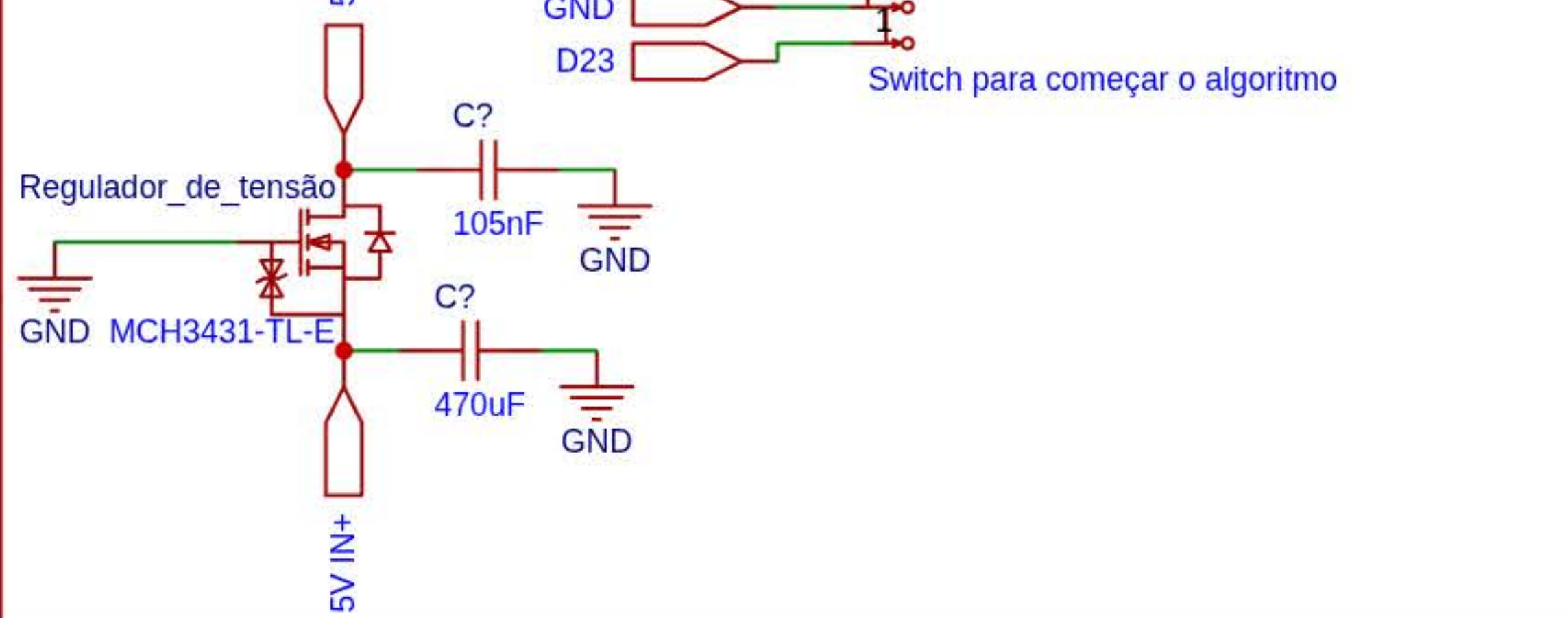
- Power:** 3.3V and 5V supplies are connected to the appropriate pins.
- Motor Control:** MA_CH1, MA_CH2, MB_CH1, and MB_CH2 are connected to the motor driver module.
- Interrupts:** INT1, INT2, INT3, and INT4 are connected to the motor driver module.
- External Devices:** SWITCH 3, SDA_INA219, Gpio0 - IV 1, Gpio0 - IV 2, Gpio0 - IV 3, and SCL_INA219 are connected to the microcontroller.

Filtro capacitivo



	Schematic	Schematic1			Create at	2026-05-10
	Board	Board1			Update at	2026-07-08
	Drawn	2026_1_P11_Esquematico_MicroMouse				
	Reviewed					
		Version	Size	Page 1 Total 1		
	V1.0	A4	EasyEDA.com			
1	2	3	4	5	6	



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